Literature Review on Machine Learning for Seizure Detection and Prediction Using Non-EEG Wearable Biosignals

Paulin Saher
Universität zu Köln

December 1, 2025

Contents

1 Introduction

1.1 Problem Statement and Motivation

Artificial intelligence (AI) and, in particular, deep learning have enabled substantial advances in medical signal analysis, supporting early detection, and continuous monitoring.

The focus of this Literature review is on newer deep learning models and other traditional machine learning for epileptic seizure detection and prediction using non-EEG wearable/autonomic biosignals (ECG/HRV, PPG, EDA, ACC). A key focus will be on how the inclusion of time series data into the context of the model, novel deep learning architectures, training strategies, and personalized fine tuning can improve performance (sensitivity, false alarms per hour (FAR/h), latency) in clinical and ambulatory contexts.

Epilepsy affects roughly 7.6 per 1,000 people and about 30% of patients are drug resistant, motivating demand for timely, low-burden monitoring and early intervention (Beghi, 2019; Chen et al., 2020; Kwan & Brodie, 2000). While EEG remains the reference in controlled settings, EEG hardware is obtrusive for pervasive use. Wearables offer the possibility of continuous monitoring in ambulatory settings (Chen et al., 2020). The key question is which model choices and architectures on wearable signals deliver clinically viable performance outside the clinic.

Model-centric challenges

- Operating point: achieving high sensitivity at acceptable FAR/h via personalized fine-tuning, calibration, thresholding, and alarm logic.
- Robustness and generalization: handling motion artifacts, physiological non-stationarity, and domain shift (patient/device/site/activity).
- Interpretability and safety: explanation and fail-safes suitable for regulated healthcare.
- Deployment context: clinic vs everyday use, sample- vs alarm-based evaluation, latency/energy limits for on-device inference, and the possibility for edge computing.
- Data limitations: scarce prospective, patient-preserving, multimodal wearable datasets and the need for external/cross-dataset validation, or real world testing.

1.2 Evaluation Protocols (brief Overview)

We distinguish the following evaluation elements:

1. **Study designs:**
   - Retrospective
   - Pseudo-prospective
   - Prospective
2. **Evaluation metrics:**
   - Sample-based metrics
   - Alarm-based metrics
3. **Data splits:**
   - Patient-preserving splits

Alarm-based sensitivity, false alarms per hour (FAR/h), and latency better reflect real-world usability than sample-based accuracy and are key performance metrics. Cross-dataset transfer is essential to assess robustness and domain shift (Andrade et al., 2024; Yang et al., 2022).

1.3 Research Aim and Questions

This literature review focuses on machine/deep learning models for seizure detection and prediction using non-EEG wearable/autonomic biosignals, and determines the optimal strategies to achieve clinically meaningful performance in ambulatory contexts.

1.3.1 Primary Research Question

In non-EEG wearable seizure detection and preictal prediction (ECG/HRV, PPG, EDA, ACC), which modeling and evaluation choices reported in 2015–2025 are associated with low alarm-based false-alarm rates and high sensitivity with acceptable latency in ambulatory settings?

1.3.2 Sub-questions

1. What ranges of sensitivity, FAR/h, and latency are reported under retrospective, pseudo-prospective, and prospective designs, and how do alarm-based results differ from sample-based metrics?

2. Across studies, how do traditional feature-based models compare to 1D-CNN/TCN/LSTM, and what evidence exists for attention/ transformer variants on these signals, including any reported computational/latency considerations?

3. How do temporal choices (window length, stride, context length) correlate with reported alarm-based performance in real-time use?

4. What evidence is available on robustness to domain shift (patient/device/site/activity), including cross-dataset/external validation and how do these affect alarm-based metrics?

5. What dataset gaps and class imbalance issues are reported, and what is the measured impact of augmentation or synthetic data on alarm-based outcomes?

Contribution. We synthesize a taxonomy of model designs for wearable seizure detection/prediction, analyze how evaluation choices affect reported performance, and derive practical recommendations to achieve low FAR/h with high sensitivity in ambulatory deployment.

1.4 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by **websterAnalyzingPastDetermine2002** and PRISMA principles). The methodological approach uses a multi-stage search and transparent extraction/synthesis procedures as outlined below.

1. **Search strategy (multi-stage, 2015–2025):**

    - Scoping: Google Scholar for broad coverage and identifying candidate papers.
    - Formal queries: IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.

- Citation chasing: backward and forward citation tracking on key papers to ensure completeness.

2. **Inclusion / Exclusion criteria:**

   - Inclusion: human studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
   - Exclusion: EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate a concept matrix with the following fields:

1. Setting: clinical vs ambulatory.
2. Modality: ECG/HRV, PPG, EDA, ACC.
3. Task: detection vs prediction.
4. Model class and fusion strategy.
5. Study phase: retrospective, pseudo-prospective, prospective.
6. Evaluation granularity: sample- vs alarm-based.
7. Metrics: sensitivity, FAR/h, AUC, latency.
8. Personalization.
9. Interpretability and safety considerations.
10. Data characteristics and dataset identity.

Quality flags: patient-preserving splits; external/ cross-dataset validation; and FAR/h reporting. A PRISMA-style flow diagram will summarize screening and inclusion. Synthesis is narrative, grouped by modality and task, with emphasis on model design levers and deployment feasibility.

2 Modeling landscape and design dimensions

2.1 Signals, tasks, and problem setup

We target detection and preictal prediction from ECG/HRV, PPG, EDA, and ACC. Windowing, labeling, and alarm logic define the operating point; alarm-based metrics (sensitivity, FAR/h, latency) are primary for deployment.

2.2 Representation learning and temporal modeling

Classical models on HRV/EDA/ACC features (e.g., SVM, RF, kNN) achieve solid detection and some prediction, often in retrospective designs (Mason et al., 2024; Seth et al., 2023). Deep architectures (1D-CNN, LSTM, TCN, transformers) improve robustness and phase discrimination, especially with attention (Abtahi et al., 2025; Kalousios et al., 2024; Pordoy et al., 2025; Yang et al., 2022).

2.3 Multimodal fusion

Feature-, decision-, and representation-level fusion can exploit complementary ECG/EDA/ACC/PPG cues; attention-based fusion often yields gains under noise and non-stationarity (Pordoy et al., 2025; Yang et al., 2022).

2.4 Personalization, calibration, and drift

Inter-patient variability degrades population models; patient-specific fine-tuning and post-hoc calibration reduce FAR/h and improve sensitivity in ambulatory, alarm-based settings (Andrade et al., 2024). Thresholding and periodic recalibration mitigate drift.

2.5 Robustness and domain shift

Cross-dataset evaluation (e.g., TUH→RPAH/EPILEPSIAE) exposes domain shift and favors alarm- over sample-based metrics (Andrade et al., 2024; Yang et al., 2022). Artifact handling, augmentation (time-warping, jitter, scaling), and OOD checks support generalization (Kalousios et al., 2024; Seth et al., 2023).

2.6 Deployment considerations

On-device inference requires latency/energy awareness; compact architectures and event-driven processing are preferred. Interpretability and alert justification are important for safety and clinical acceptance.

3 Technical background (clinical and sensing context)

3.1 Epileptic seizures (brief Overview)

Seizures are classified by onset as focal (aware vs impaired awareness; motor vs non-motor, may evolve to bilateral tonic–clonic) and generalized (motor: tonic–clonic, myoclonic, atonic; non-motor: absence) (Fisher et al., 2017). Tonic–clonic seizures commonly cause loss of consciousness and carry increased injury/SUDEP risk (Hesdorffer et al., 2011).

3.2 Autonomic markers and wearable sensing

Sympathetic activation around seizures manifests as ictal tachycardia, preictal HR rise with reduced HRV (lower RMSSD/HF, higher LF/HF), increased EDA (tonic SCL, phasic SCRs), and limited ACC changes useful for artifacts/context. PPG yields pulse-rate variability analogous to HRV from ECG (Mason et al., 2024; Miron et al., 2025). Wearable heart-rate sensing aligns closely with chest-strap ECG under motion (e.g., Apple Watch Ultra/Polar H10 concordance ~0.998), though dominant-wrist motion degrades accuracy; chest-strap ECG provides robust RR intervals during exercise (Wolf et al., 2024).

4 Methodology

1. **Scope:** human studies in journals (2015–2025) on non-EEG wearable/autonomic biosignals (ECG/HRV, PPG, EDA, ACC) for seizure detection or prediction.

2. **Search:** Google Scholar (scoping), IEEE Xplore (all queries), PubMed and Scopus (targeted) using predefined query strings and deduplication.

3. **Screening:** title/abstract screen, then full-text eligibility; backward/forward citation chasing on key papers.

4. **Extraction:** dataset type (wearable vs clinical), name, availability, task (detection vs prediction), evaluation design (retrospective/pseudo-prospective/prospective), metrics (sensitivity, FAR/h, AUC, latency), personalization, cohort size/ class balance.

5. **Quality flags:** sample- vs alarm-based evaluation; patient-preserving splits; cross-dataset/externa validation.

6. **Synthesis:** narrative grouping by modality and task with emphasis on modeling design levers and ambulatory feasibility.

5 Key Research Questions

1. Which wearable biosignals (ECG/HRV, EDA, ACC, PPG) and model classes (feature-based, 1D-CNN/LSTM/TCN, transformers, multimodal fusion) maintain performance under pseudo-/prospective, alarm-based evaluation?

2. What performance gains arise from patient-specific adaptation, and how can calibration/fine-tunin be implemented under edge constraints?

3. How can FAR/h be minimized via thresholding, calibration, and alarm logic while preserving high sensitivity in ambulatory use?

4. How robust are models to domain shift (patient/device/site), and what is the impact of cross-dataset/external validation and OOD checks?

5. What dataset gaps exist for non-EEG wearables, and how can augmentation/synthetic data mitigate imbalance without bias?

6 Overview of Literature Review

This review is organized as follows.

1. **Modeling landscape:** signals and tasks; representation learning; temporal modeling; fusion; personalization; robustness; deployment.

2. **Clinical/sensing context (brief Overview):** seizure taxonomy; ANS markers; wearable sensing accuracy.

3. **Methodology:** search, selection, extraction; PRISMA flowgraph.

4. **Results:** concept matrix; comparative performance; modality/model analysis.

5. **Discussion:** interpretation, trade-offs, safety, deployment.

6. **Limitations and future work.**

7. **Conclusion.**

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